THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Control and Coordination** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An impulse is blocked in the sensory neuron only. The patient will: 1

(a) not feel a stimulus **(b)** feel but not move
(c) move more **(d)** have no reflex only

2. A person has sudden weight gain, intolerance to cold and a slow heart rate. The gland to be checked is the: 1

(a) thyroid **(b)** adrenal **(c)** pituitary only **(d)** pancreas

3. Assertion (A): The pancreas is both an exocrine and an endocrine gland. Reason (R): It secretes digestive juice and insulin. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Plants lack nervous system, yet they respond to stimuli by: 1

(a) hormones and turgor changes **(b)** blood
(c) muscles **(d)** neurons

5. After eating a heavy meal, the secretion of which hormone increases? 1

(a) adrenaline **(b)** insulin **(c)** thyroxine **(d)** growth hormone

6. Assertion (A): Movements of the sensitive plant are due to growth. Reason (R): The movement is caused by a change in turgor pressure of cells. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A plant is placed horizontally. After some days the shoot bends up and root bends down. This is due to: 1

(a) thigmotropism **(b)** geotropism **(c)** hydrotropism **(d)** phototropism

8. A runner's reaction at the start of a race is due to: 1

(a) insulin release **(b)** less thyroxine
(c) adrenaline release **(d)** growth hormone

9. **Assertion (A):** Insulin is secreted by the pancreas. 1
Reason (R): Insulin raises the blood sugar level.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Adrenaline is secreted by the pituitary gland. 1
Reason (R): Adrenaline prepares the body to face an emergency situation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If a signal travels at 100 m/s along a nerve, how long does it take to travel 1.5 m to the brain? 2

12. Why does the heart rate increase during exercise even without adrenaline? 2

13. Why do the stems of a plant kept on the windowsill bend toward the window? 2

14. A child lacks iodine in diet for years. Name two effects. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why does pancreatic damage lead to an unusual rise in blood sugar and what changes in urine? 3

16. Explain why the reaction time to a touch on the toe is longer than for a touch on the face. 3

17. Explain how growth in a plant stem differs from the movement in a sensitive plant leaf. 3

OR

What would happen if the thyroid gland is removed? Explain with the role of iodine. 3

18. Compare the roles of auxin and cytokinin in plant growth. 3

OR

State two reasons why hormones are not given in food as medicines (like insulin). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Mr Gupta's blood test shows a high sugar level. His doctor says that a hormone is not being secreted in sufficient quantity by a gland that lies below the stomach. Mr Gupta's cousin, who lives in a hilly village and does not use iodised salt, has a swollen neck. 5
(i) Name the hormone lacking in Mr Gupta.
(ii) Name the gland that secretes it.
(iii) Name the disease of the cousin's neck.
(iv) Name the hormone whose synthesis is affected in the cousin and state one function of that hormone.

20. A diabetic and a non-diabetic person eat the same meal. Compare blood sugar and insulin over 3 hours, and explain. 5

OR

Describe five disorders linked with hormones and the cause, symptoms and prevention or treatment. 5

END OF QUESTION PAPER — Total: 40 Marks